Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/15/2021
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
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City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per C. Ribeiro/JSC and NSSC information (Ed., 4/7/22) NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21) NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21) NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	80NSSC18K0310		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	The objectives of this project are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species are available for sharing from three NASA funded investigators. We will evaluate radiation-induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures. Archived ocular tissues from single and 12 equal dose fractions irradiated mice have received from collaborators in mid-2020. At four months following irradiation, mice were sacrificed and ocular tissues were collected for histochemical analysis in the retina and optic nerve head region (ONH). Fixed ocular samples were sectioned for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina samples were used for genomic pathway analysis. Significant increases in the number of apoptotic cells were documented in the mouse retinas and ONHs that received proton radiation with a single or fractionated dose ($p<0.05$). Immunochemical analysis revealed enhanced immunoreactivity for oxidative biomarker, 4-hydroxynonenal (4-HNE) in the retina and ONH following single or fractionated protons with more pronounced changes observed with a single dose of 0.5 Gy. Our study also revealed the differences in BRB biomarker distribution between these two regions. In response to radiation insults, the cellular response in the retina and ONH may be differentially regulated in acute or hyperfractionated dose schedules. Our data was accepted for oral presentation at the American Society for Gravitational and Space Research (ASGSR) 2021 meeting. Our data is also scheduled to be presented at International Astronautical Congress (IAC) in November, 2021. One paper titled: Evaluating ocular response in the retina and optic nerve head after single and fractionated low-dose high-energy protons is published in Life (Basel). 2021 Aug 19; 11(8):849. PMID: 34440593. Augmentation fund is awarded to evaluate mouse retina and optic nerve head changes following spaceflight and elucidate the impact of artificial gravity (AG) in mitigating the effects of microgravity on board International Space Station (ISS) on ocular structure and BRB integrity using mouse Habitat Cage Unit (HCU) in the Japan Aerospace Exploration Agency (JAXA) ‘Kibo’ facility in the ISS. The flight study is currently scheduled in October, 2022.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Delp M. "Lunar Spaceflight-Induced Effects on Ocular Response and Blood-Retina Barrier Function." Lunar Surface Science Workshop Space Biology, Virtual, January 20–21, 2021. Abstracts. Lunar Surface Science Workshop Space Biology, January 20–21, 2021. , Jan-2021

Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391; PMCID: PMC8274610 , Apr-2021
Articles in Peer-reviewed Journals	Mao XW, Stanbouly S, Jones T, Nelson G. "Evaluating ocular response in the retina and optic nerve head after single and fractionated low-dose high-energy protons." Life (Basel). 2021 Aug 19;11(8):849. https://doi.org/10.3390/life11080849 ; PMID: 34440593; PMCID: PMC8400407 , Aug-2021
Articles in Peer-reviewed Journals	Paul AM, Overbey EG, da Silveira WA, Szewczyk N, Nishiyama NC, Pecaut MJ, Anand S, Galazka JM, Mao XW. "Immunological and hematological outcomes following protracted low dose/low dose rate ionizing radiation and simulated microgravity." Sci Rep. 2021 Jun 1;11(1):11452. https://doi.org/10.1038/s41598-021-90439-5 ; PMID: 34075076; PMCID: PMC8169688 , Jun-2021